Retail Profit Leakage Analysis

An E-commerce Sales and Profitability Analysis Using SQL (SSMS) and Power BI

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Tools used: Microsoft SQL Server Management Studio (SSMS) and Microsoft Power BI

1. Executive Summary

This project looks at 8 sets of backend data from an online retail business (sales, customers, returns, products, pricing, stock, supplier costs, and ad spend) to answer one simple question: **"is this business really making as much money as it looks like it is, and if not, where is the money going"?**

Using SQL to clean and combine the data, and Power BI to turn it into charts, the project builds a "real profit" number, one that also subtracts refunds and ad spend, not just the cost of the goods themselves. The gap between what the business *reports* as profit and what it *actually* keeps is the main story of this report.

What I found:

- Real profit is 14–16 percentage points lower than reported profit, and that gap has been getting bigger every year from 2022 to 2024.
- Running promotions cuts profit margin by over 12 points without actually selling more, discounts aren't earning their keep.
- Paid Search ads are the worst-performing ad channel (only $6.30 back for every $1 spent) despite getting the most money, while free traffic (Direct and Organic Search) performs far better for almost nothing.
- Three fixable reasons, wrong item descriptions, broken products, and wrong items sent, make up about $2.56M in refunds, more than half of all return costs.
- Customers aged 55 and older bring in almost double the profit of any other age group, suggesting the business may be marketing to the wrong people.
- No single product actually loses money, but the Books category earns far less than Electronics or Clothing.

What I recommend: Cut back on blanket discounts, move ad money away from Paid Search toward cheaper channels, fix the reasons behind the top 3 return problems, restock items about to run out, and make sure ads are actually reaching the 55+ customers who spend the most. Together, these 5 changes could recover an estimated **$750,000–$1,000,000+** a year.

2. Introduction and Business Problem

Most online stores only look at their sales numbers: how much sold, and what the "profit" says. But that misses most of what actually decides whether a business is doing well. A sale that looks profitable on paper can still cost the business money once you count what it took to get that customer (ads) and what it costs when they send it back (refunds). Stock sitting unsold ties up cash, and stock that runs out quietly loses sales nobody ever records.

This dataset connects every part of the business, sales, customers, pricing, stock, suppliers, and ad spend, using shared ID numbers. That means we can go beyond "what sold" and ask "what did it really cost us to sell it, and was it worth it?"

The problem: This retailer is growing revenue, but is it growing profit? This project traces where money is made and where it quietly disappears, across marketing, pricing, returns, and stock, to show what the business should change.

This matters because a business that only watches its "reported profit" number can feel like it's doing great while actually losing ground, wasting money on ads that don't work, discounting things that didn't need it, or quietly absorbing return costs it could have avoided. Finding these leaks is what turns raw sales data into a real action plan for management.

3. Project Objectives and Key Questions

The goal is to work out the business's real profit and point to specific, costed ways to improve it, across marketing, products/pricing, customers, and operations (returns and stock).

Questions we set out to answer:

1. How much does the business really make each month, once refunds and ad spend are counted?
2. Which products make the most money, and which make the least?
3. Which ad channels are worth the money, and which are losing it?
4. Do premium customers actually spend more than regular customers?
5. Which countries and age groups bring in the most profit?
6. Does our price compared to competitors affect how much we sell?
7. Do promotions actually raise profit, or just move more product at a lower margin?
8. What gets returned, why, and how much does it cost?
9. Which products are about to run out, and which have too much stock?
10. What should management do first, and how much is each action worth?

4. Dataset Description

The data is a simulated set of 8 CSV files for an online store, loaded into a SQL Server database. The tables connect to each other using shared ID columns: product ID, customer ID, transaction ID, and year-month.

Table	What's in it	Main columns
transactions	Every sale: date, quantity, discount, revenue, cost, profit, payment method, channel, status	transaction ID, customer ID, product ID, date
customers	Age, country, sign-up date, premium status	customer ID, country, age, is premium
returns	Returns: reason, refund amount, whether restocked	return ID, transaction ID, product ID
products	The product catalog: category, brand, weight, launch date	product ID, category, brand

Table	What's in it	Main columns
price_history	Monthly pricing: our price, competitor price, promo status, units sold, margin	product ID, year-month
inventory	Warehouse stock: stock level, reorder point, lead time	product ID, warehouse location
supplier_costs	What suppliers charge: unit cost, ordering cost, holding cost, minimum order size, reliability	product ID, supplier name
marketing_spend	Monthly ad spend by channel: impressions, clicks, cost per customer, return on ad spend	year-month, channel

Time covered: January 2022 to December 2024 (3 full years of monthly sales data).

Main fields used throughout: revenue, profit, refund amount, ad spend, promo flag, competitor price, stock units, reorder point.

What's missing or limited:

- Ad spend is only recorded by month and channel, not tied to a specific customer, so we can see which channels work overall, but not which exact customer came from which ad.
- This is simulated data, so some patterns (like how price affects sales) may be weaker or cleaner than they'd be in a real business.
- The data doesn't give us everything needed for a full reorder-quantity formula, so stock recommendations use a simpler "how many days of stock is left" approach instead.

5. Data Cleaning and Preparation

All the work was done in SQL Server Management Studio using SQL.

Wrong data types. When the data was first loaded, every single column in all 8 tables was stored as plain text, including dates, prices, and quantities. That meant SQL couldn't sort dates properly or add up money until it was fixed. Each column was converted to its correct type (date, number, true/false), with a safety check first to make sure nothing would get lost or turned into a blank value by accident.

Missing values. Every column in every table was checked for blanks, both truly empty cells and text that looks empty (like "NULL" or "NaN"). None were found.

Duplicates. Every table was checked to make sure no transaction, return, or product ID appeared more than once where it shouldn't. None were found.

Broken links between tables. Every connection between tables (sales linking to customers, returns linking to sales, stock linking to products, etc.) was checked to make sure nothing pointed to an ID that didn't actually exist. Nothing was broken.

Inconsistent formats. The year-month column wasn't always written the same way across tables, so it was standardized to one format so the tables could be joined correctly.

New calculated fields. A few extra columns were built to support the analysis:

- Year-month pulled out of each sale's date, for monthly totals

- "Real profit" = reported profit minus refunds minus ad spend
- Price position (cheaper / about the same / more expensive than competitor)
- Age groups (under 25, 25-34, 35-44, 45-54, 55+)
- Return rate % and days of stock left
- A stock status flag (reorder now / will run out / too much stock / fine)

Unusual values. Nothing was deleted just for looking unusual. Instead, anything strange (like one customer with an unusually high number of orders) was flagged and written up as a finding, rather than removed, since deleting it without checking first could accidentally hide a real, legitimate high-value customer.

6. Exploratory Data Analysis and Methodology

The analysis was done in SQL for combining and totaling the data, then turned into visuals in Power BI. 23 core queries were written, each one answering a piece of the 10 business questions.

How it was done:

- **Staged calculations (CTEs):** monthly sales, monthly refunds, and monthly ad spend were calculated separately first, then joined together into one "real profit" view.
- **Grouping and totaling:** category, channel, country, age group, and product totals were built using simple grouping and summing.
- **Safe joins:** tables were combined carefully so that a sale with no matching return wouldn't get lost or cause a divide-by-zero error.
- **Grouping rules:** ages were split into 5 bands, prices into 3 competitive positions, and stock levels into 4 risk categories, using simple if-this-then-that logic.
- **Percentages and ratios:** margin %, return rate %, return on ad spend, and cost per customer were all calculated directly in SQL.
- **Double-checking:** figures already in the data (like units sold) were compared against totals built from scratch, to confirm the data was internally consistent.

Numbers we tracked throughout:

- Revenue, reported profit, real profit, and profit margin %
- Return on ad spend and profit-after-ads by channel
- Return rate % and refund $ by category and reason
- Revenue per customer (premium vs non-premium)
- Days of stock left and stock status by product

How the dashboard was built: Power BI was laid out as a 4-page story, Overview, Products, Marketing, Customers & Pricing, and Summary, with the chart type matched to the question: line charts for trends over time, bar charts for rankings (products, channels, customers), a waterfall chart to show money disappearing step by step, donut charts for small categories (return reasons, price position, promo status), and tables with color flags for action lists (low stock). A country filter was linked across every page so filtering by country updates the whole dashboard at once.

7. Results and Key Findings

For each finding below: **What we found**, **Why it matters**, **What to do about it**.

7.1 Real profit is well below reported profit, and the gap is growing

What we found: Looking at December each year, reported profit margin stayed about the same (around 34%), but real profit margin (after refunds and ad spend) dropped from 20.3% in 2022 to 19.9% in 2023 to 18.8% in 2024. By late 2024, almost 45% of reported profit was being eaten up by refunds and ads, up from 41% two years before.

Why it matters: The business looks just as healthy every year on paper, but it's actually making less money underneath. Watching only reported profit would hide this.

What to do: Start tracking real profit margin, not reported profit margin, as the main number that matters, and tackle the two things driving the gap: rising ad costs and return costs (below).

7.2 Marketing is getting less efficient

What we found: Ad spend as a share of revenue went up every year: 7.0% in Jan 2022, 7.9% in Jan 2023, 9.0% in Jan 2024.

Why it matters: The business is spending more and more just to make the same sale. This is a big reason real profit keeps shrinking.

What to do: Track ad spend as a % of revenue as its own number, and fix the channel problems below.

7.3 Bigger sales don't always mean bigger profit

What we found: One product (Electronics) brought in the most revenue ($1.99M) but only kept 20.3% of it as profit. Another product (Clothing) brought in 62% less revenue ($1.23M) but kept 52.4% as profit, and ended up with more total profit.

Why it matters: If the business only chases revenue, it ends up over-investing in products that don't actually make much money.

What to do: Judge products by profit, not revenue, when deciding what to promote or restock.

7.4 Paid Search is the worst ad channel, despite getting the most money

What we found: Paid Search spent $1.86M (the most of any channel) but only returned $6.30 for every $1 spent, and ended up with $2.09M profit after ad costs. Direct traffic spent just $40,884 and returned $130 for every $1 spent. Organic Search spent $212,115 and made $4.8M profit after ad costs, the best of any channel.

Why it matters: The two cheapest channels made $6.5M combined for only $253,000 spent. Paid Search and Social Media together cost $3.1M and made about $4.08M combined. The business is spending over 12 times more to get a similar result.

What to do: Cut Paid Search and Social Media spend by about 20–30%, and put that money into SEO and keeping existing customers instead.

7.5 Promotions cut profit without selling more

What we found: Products on promotion sold slightly fewer units on average (12.6 vs 12.9) and kept only 33.8% margin, compared to 46.2% for products not on promotion.

Why it matters: Promotions are giving away over 12 points of profit margin without even selling more to make up for it.

What to do: Stop running blanket discounts. If promotions continue, only use them to clear slow-moving stock, and check they actually raise profit, not just sales.

7.6 Price compared to competitors barely affects what sells

What we found: Whether a product was priced cheaper, the same, or more expensive than competitors didn't show a strong, consistent effect on how many units sold.

Why it matters: This corrects an assumption, price position isn't the main problem here. The real issues are promotions (7.5) and ad spend (7.4).

What to do: Spend less effort chasing competitor prices, and focus on protecting margin instead.

7.7 Premium customers aren't really worth more

What we found: Non-premium customers spent $6,606.84 on average; premium customers spent $6,704.07, only 1.5% more.

Why it matters: The premium membership isn't driving meaningfully more spending right now. That's surprising, since premium usually means much higher value.

What to do: Review what premium membership actually offers and whether it needs to be redesigned to earn its name.

7.8 Margins are steady everywhere; the real opportunity is reaching more people

What we found: Profit margin stayed between 33.4% and 35.0% across every country, even though revenue ranged from $1.09M (Sweden) to $16.27M (USA). Margins were also steady across age groups, but customers 55+ brought in $6.38M in profit, almost double the next group.

Why it matters: Since margins are already healthy everywhere, growth should come from reaching more customers, not changing prices. The strong 55+ result is unusual and may not match where current ads are aimed.

What to do: Grow marketing in smaller but equally profitable countries (Sweden, Netherlands, Australia), and check whether ads are actually reaching the 55+ customers who already spend the most.

7.9 Three fixable reasons cause most of the return costs

What we found: Out of 5 return reasons, "not as described" ($894,772), "defective" ($931,700), and "wrong item" ($738,439) add up to $2.56M, over half of all refund money.

Why it matters: All three are fixable: wrong descriptions and photos, poor quality checks, and packing mistakes, not customers simply changing their minds.

What to do: Fix product listings (especially Electronics, see below), tighten supplier quality checks, and double-check orders before shipping. Fixing even 20% of these could save about $512,000.

7.10 Return rate looks similar everywhere, but the money lost isn't

What we found: Return rates were close across all categories (6.87%–7.31%), but Electronics, despite one of the lowest rates, cost $1.7M in refunds, more than every other category combined.

Why it matters: Return rate alone is misleading. Because Electronics items cost more, even an average return rate means a much bigger dollar loss.

What to do: Focus return-fixing efforts on Electronics first, it has the biggest payoff.

7.11 A lot of products are about to run out

What we found: Many products are flagged "reorder now" (very low stock) or "will run out before restock" (stock will be gone before the next delivery arrives).

Why it matters: These aren't future risks, they're guaranteed to run out under current ordering, which means guaranteed lost sales.

What to do: Reorder every "will run out before restock" product right away, and set up automatic reorder alerts going forward.

7.12 No product actually loses money, but Books underperforms

What we found: Every single product made some profit overall, none were negative. But 6 of the 10 weakest products were Books, some earning as little as $584–$1,621 total, compared to top products earning $150,000–$645,000.

Why it matters: This corrects an assumption too, there's nothing to "cut" outright. The real opportunity is where to put marketing and shelf space.

What to do: Shift marketing and shelf space away from Books and toward Electronics and Clothing, rather than discontinuing specific products.

8. Insights and Recommendations

Pulling all 12 findings together, here are the 5 biggest, most doable next steps for management.

#	What to do	Based on	What it's worth
1	Stop blanket discounts; only promote narrowly	7.5	~$230,000 back in margin
2	Move ad money from Paid Search/Social to Organic Search and retention	7.4	The biggest opportunity of all 5
3	Fix the 3 biggest return reasons (listings, quality checks, packing)	7.9, 7.10	~$512,000 saved in refunds
4	Reorder every product about to run out, right away	7.11	Stops sales being lost for no reason
5	Make sure ads reach the 55+ / USA-UK-Germany customers who already spend the most	7.4, 7.8	Protects and grows the most valuable, most loyal customers

Together, these 5 changes could recover roughly $750,000–$1,000,000+ a year, without needing new products, new markets, or new customers, just fixing what's already going wrong.

One honest note: Finding 7.12 showed that no single product is actually losing money, so the fix there is about moving investment (less on Books, more on Electronics/Clothing), not cutting products outright. Including a finding that corrected our own first guess, instead of ignoring it, is part of what makes this analysis trustworthy.

9. Conclusion

This project set out to answer one question: is a growing online store actually making more money, or just moving more product? By combining 8 sets of backend data in SQL and building a "real profit" number that subtracts refunds and ad spend, the answer is clear: the gap between what looks like profit and what's actually kept is real, significant (14–16 margin points), and getting worse every year.

The most important discoveries weren't just that money is leaking, it's exactly where: a promotion strategy giving away margin for nothing, ad money going to the worst-performing channel, return problems that are fixable, and a mismatch between where the most valuable customers (55+, consistently strong everywhere) actually are and where ads are probably being aimed.

Every finding was checked against the real numbers instead of assumed, including one case (7.12) where the first guess, that some products were losing money outright, turned out to be wrong, and the recommendation was changed to match what the data actually showed. That's the difference between a real data-driven case and a story built to fit a conclusion.

Put together, the 5 recommendations in Section 8 directly answer the original question: they show exactly where the money is leaking, put a dollar figure on each one, and give management a clear, evidence-backed plan, worth an estimated $750,000–$1,000,000+ a year, achievable without needing new products, new markets, or new customers.